



SUPERWOOD

SUPERWOOD for Data Centers

Companion to Super Mills America · September 2026

INVENTWOOD · CONFIDENTIAL

What's inside

The company, the mills, the cost roadmap and the raise are in Super Mills America. This deck covers one application: data centers.

01

Why Data Centers

Hungry for materials · large offtake · low carbon and community acceptance · the wedge into structural

02

Technology Fit

How SUPERWOOD is made, what it does, and the standards on the path

03

Impact

Embodied carbon and circularity — labeled as the projections they are

04

Applications

Skins and fences now; certification-gated next; structure after

05

Customers

Microsoft, Meta, Google — and who else is at the table



SUPERWOOD boards

SECTION 01

Why *Data Centers*

Fast-growing and hungry for materials · large offtake · low carbon and community acceptance · the wedge into structural

Why *data centers*?

01

Fast-growing market, hungry for materials

Construction is backlogged and operators tell us they face a shortage of structural steel. Supply chain and timeline are their first concern — and a domestic material that is lighter to ship and faster to erect answers it directly.

02

Large-volume offtake potential

Hyperscalers build repeatable campuses to a standard basis of design. One campus envelope program is on the order of SuperMill One's entire annual output; a basis-of-design win is SuperMill Two-scale demand.

03

Desire for low carbon and community acceptance

Microsoft and Meta already build data-center structures with mass timber (35% and ~41% embodied-carbon cuts on the elements replaced), reported through EC3 — co-created by our advisor Don Davies. Noise and visual mass drive moratoria; warm, quiet, biophilic exteriors change what the planning commission sees.

04

A wedge into structural — a market hungry for solutions

Replacing structural steel is the first thing new customers ask about. The mass-timber code pathway lets SUPERWOOD qualify as a new, far stronger species — and what is qualified for a data center is qualified for the broader structural market.

HOW THE WEDGE WORKS

1

Skins and fences today

2

Testing, co-developed with customers

3

Structural qualification via the mass-timber pathway

4

Written into the basis of design

5

SuperMill Two-scale offtake

Steel-shortage and backlog statements are what customers report to us (2026). Microsoft: news.microsoft.com, Nov 2024. Meta: sustainability.atmeta.com, Jul 2025. Campus-vs-SuperMill One comparison is a company estimate.

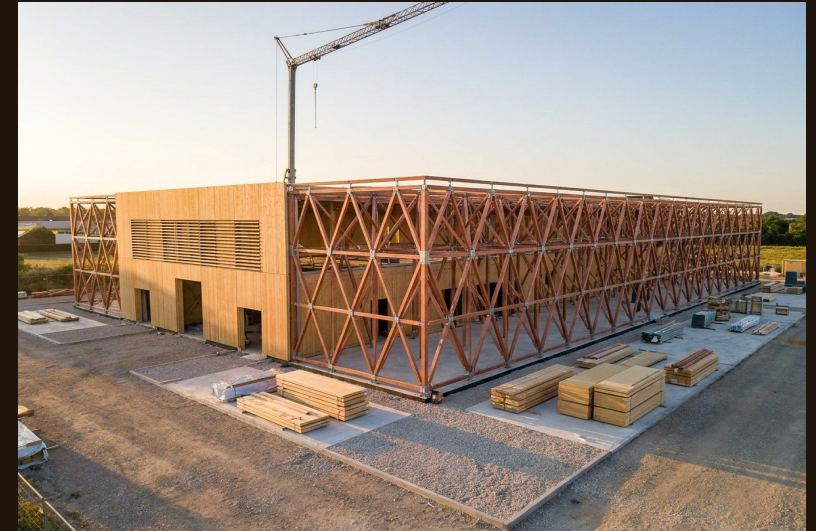
Where SUPERWOOD fits in a campus

NOW · SKINS, SCREENS, FENCES

- Facades, cladding and rain screens
- Biophilic interiors for office space
- Louvers and equipment screening
- Fencing for staff outdoor spaces
- Security fencing around transformers and outdoor infrastructure
- Equipment backplanes
- Trim, door kicks, sub-framing

NEXT · STRUCTURE AND CONTENTS

- Building enclosures and structural components
- Joists, deck and roof assemblies
- Trusses and long-span members
- Platforms, walkways and mezzanines
- Racking and equipment supports
- Enclosures and separations
- Shells and foundations, longer term



Concept rendering — enclosure and structure are the long game; skins and fences ship now.

Per 10 MW of data center: 500–1,000 t of structural steel and 5,000–10,000 m³ of concrete.

Steel and concrete intensities: arXiv 2509.21312 (Sep 2025), citing Hasan et al. 2022 and Sharma et al. 2023 — secondary literature figures, ranges as published.



SECTION 02

The *Technology*

How SUPERWOOD is made, what it does, and the standards on the path

We unlock the power of *cellulose*

Our patented process transforms ordinary wood into SUPERWOOD — an environmentally benign process using chemistry common to food and pulp processing.

01 · START WITH WOODY FEEDSTOCK

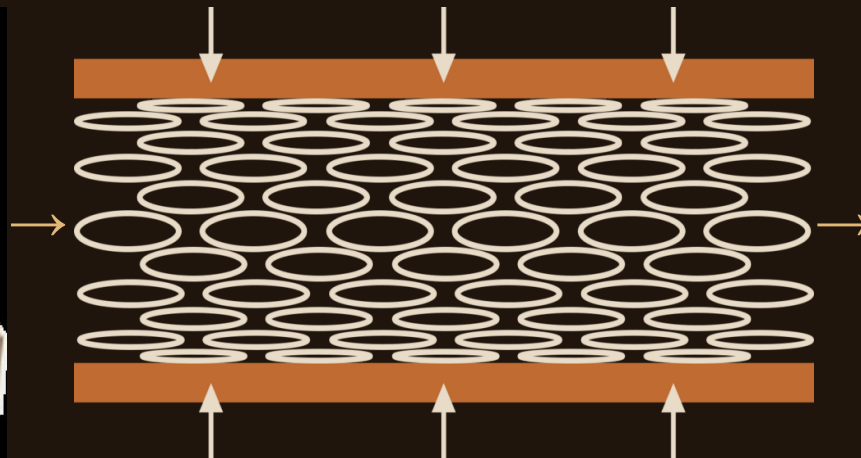
Hardwoods, softwoods, bamboo, underutilized species, waste wood



Before: open cell vessels

02 · MOLECULAR RE-ENGINEERING & DENSIFICATION

3–4× density increase, elimination of pores and defects, and new hydrogen bonds across fibers



No added glues · No polymer binders

03 · SUPERWOOD

Samples more than 50% stronger than A36 steel in tension, at one-sixth the weight



After: collapsed, interlocking vessels

The *strength* of SUPERWOOD

50%+

stronger than A36 steel in tension —
demonstrated in samples

1/6

the weight of steel

Up to 10×

strength-to-weight ratio of steel

600 MPa

in production today; 1,000 MPa
demonstrated in the lab

SUPERWOOD: parallel-to-grain tension, production material. Steel: ASTM A36 structural. Aluminum: 6061-T6. Company test data; design values and methods available on request.

Properties that matter on a data center campus



**Stronger than steel.
One-sixth the weight.
Made from wood, in
America.**



FIRE RESISTANT

ASTM E84 Class A (surface flame spread) and aerospace standards; dense natural char layer. Assembly ratings: see roadmap



DURABLE AND IMPACT RESISTANT

3× more dent-resistant than oak; impact and storm resistant — suited to fences, screens and door protection



RESISTS MOISTURE, PESTS AND ROT

No rust; protects against termites, mold and fungus for exterior and yard use



THERMALLY AND ELECTRICALLY INSULATING

100× better than steel, 400× better than aluminum; non-conductive around electrical infrastructure



SOUND ATTENUATING

Improved damping vs. steel — the basis for acoustic barriers and equipment screens (STC/OITC testing on the roadmap)



RF TRANSPARENT

Useful in niches: antenna screening, timing-antenna enclosures, telecom shelters



NATURALLY BEAUTIFUL

The texture, colors and warmth of wood — the biophilic, community-facing surface

Performance climbs; the code pathway is mapped

	IMMEDIATE (2025–26)	MEDIUM TERM (2027–29)	LONG TERM (2030+)
Tensile strength — production	500 MPa	↑	toward 1,000 MPa
Elastic modulus — production	40 GPa	↑	toward 80 GPa
Product forms	Boards up to 8" × 16' × 3/8"	Boards + veneers (SuperMill Two)	Panels & OSB-type products (ChipMill)
Fire & code pathway	ASTM E84 Class A tested. Backplanes: UL 94 (yellow card) — testing not yet started	Trusses: no fire-resistance requirement. ASTM E119 assemblies and NFPA 285 exterior walls — not yet started	FM acceptance and ICC-ES evaluation via the mass-timber qualification pathway

Lab samples have demonstrated 600+ MPa tensile strength, with pathways to 1,000 MPa and 80 GPa. Certification programs are scoped per application with each customer; none has started yet. Test methods and property data available on request.

SECTION 03

The *Impact*

Embodied carbon and circularity — labeled as the projections they are

Can the world's data centers be a *carbon sink*?

Data centers are poised for massive growth. We asked: can we build them as long-term stores of carbon rather than massive emitters — and earn the social license to operate and scale? We think this is the way to do it:

POWERED by carbon-free electricity

Maximize efficiency and flexibility of load, then power with renewable energy, nuclear, biomass, and potentially BECCS — bio-energy with carbon capture and storage. Serve the grid.

CONSTRUCTED from waste biomaterials

Replace today's carbon-intensive materials with carbon-storing materials — designed for reuse at end of life to maximize lifecycle carbon benefit.

Steel is the lever, SUPERWOOD is the enabler. Steel sits at nearly every level of a data center — frame, joists and deck, platforms, racking, fences and screens. Replacing it element by element with a carbon-storing material is how the building's embodied ledger moves.

Manufacturing: Steel vs SUPERWOOD

Raw Material

10+ years to develop a mine; highly polluting

Universal, abundant, regenerative

Manufacturing

Energy intensive (1,800° C)
Capital intensive

Energy efficient (< 180° C)
Capital efficient

GHG Emissions — projected

1.8 kg

CO₂e per kg of steel (global average)

0.5 kg

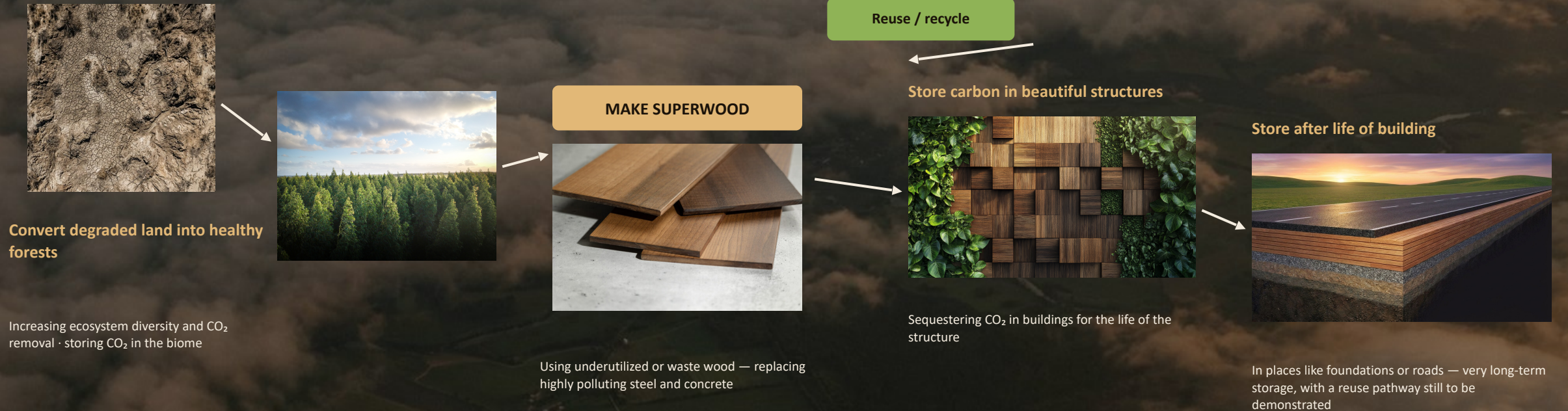
CO₂e per kg SUPERWOOD made;
1.3 kg/kg biogenic carbon stored

Pre-LCA projections for a full-scale plant; LCA under way with Prof. Ming Hu, University of Notre Dame. Biogenic storage reported separately (released at end of life under EN 15804). Recycled (EAF) steel emits less than the average shown.

STEEL

SUPERWOOD

We create an economic engine powering a healthier planet — with long-term storage in forests, in buildings, and even after the end of life of a building



Flexible, renewable, highly scalable feedstock

12.5 million tons per year of SUPERWOOD could displace 50% of steel demand in the US — roughly all the steel used in construction.

EXAMPLES OF EXCESS SUPPLY IN THE US

167 Mt/yr

Forest overburden in Western National Forests

14 Mt/yr

Excess capacity in Southern lumber mills

12 Mt/yr

Wood waste sent to landfills

2.5 Mt/yr

End-of-life fruit and nut trees

Sources: MDPI Mass Timber Consumption Report; EPA.gov; US Forest Service Sawmill Capacity and Production Report; New Leaf Management Ltd. Consulting.



SECTION 04

Data Center *Applications*

Skins and fences now; certification-gated next; structure after

Fast, cost-effective, carbon-storing data centers

01 **Faster, easier builds**

Up to 10× the strength-to-weight ratio of steel — lighter to ship and lift, faster to assemble and disassemble, with a short domestic supply chain.

02 **Safer construction and operations**

Steel is heavy, sharp, and conductive. SUPERWOOD is one-sixth the weight, without sharp edges, and electrically non-conductive.

03 **Biophilic, community-friendly design**

Natural beauty and texture that meets industrial-grade demands — a way to humanize hyperscale campuses facing public opposition, inside and out.

04 **Lower embodied carbon per element**

A bio-based material that stores biogenic carbon and is projected to emit far less in production than steel — verified element by element as the LCA lands.



Concept rendering — the street-facing view of a campus clad and screened in SUPERWOOD.

Available now — skins, screens, and fences

Every item below ships from SuperMill One today as boards up to 8" × 16' × 3/8", exterior or interior grade.



Facades, cladding & rain screens



Biophilic interiors for office spaces



Louvers & equipment screening



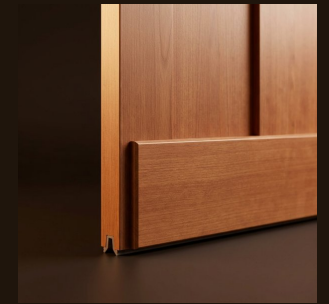
Fencing for staff outdoor spaces



Security fencing around transformers & outdoor infrastructure



Equipment backplanes



Trim, door kicks & sub-framing

Why these first. They need no assembly fire rating, they sit where the community and the workforce see the campus, and they are the near-term projects our hyperscaler customers have asked for: building facades, biophilic office interiors, staff-area fencing, and security fencing around critical outdoor infrastructure.

Concept renderings

Next — applications gated by a specific test or listing



Acoustic barrier walls & equipment screens

GATED BY

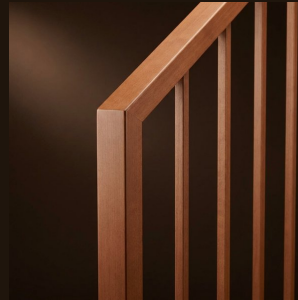
STC / OITC lab data, field insertion loss



Walkways & platforms

GATED BY

Published design values, connection data



Railings

GATED BY

Load testing to code (IBC 1607)



Window mullions

GATED BY

Thermal and structural performance data



Interior doors & door protection

GATED BY

UL 10C listing for rated openings



Industrial-grade flooring

GATED BY

Rolling-load and slip testing

Each of these is a scoped test program, not a research question — and the natural first co-funded project with a customer. Timing depends on which one a customer picks up first.

Concept renderings

Then — structure, through the mass-timber pathway



Building enclosures & structural components

The co-development target with our hyperscaler customers' architects and engineers; ultimately optimized shells and foundations



Trusses and CLT-type floor, roof & wall assemblies

Trusses carry no fire-resistance requirement; assemblies follow E119 and the PRG 320-style qualification route



HVAC enclosures & separations

Insulating, lightweight, vibration-damping enclosures and air separations — not in-airstream ducting



Racking & equipment supports

Backplanes first (UL 94), then racking and heavy-load supports as design values and seismic qualification complete

Concept renderings · SUPERWOOD can be qualified as a new, far stronger species through the pathway mass timber has already opened in the building codes. This is the structural-steel replacement our customers are asking about, and it is SuperMill Two-scale demand.

SECTION 05

Our *Customers*

Who is already building with us — engagement as of September 2026

Hyperscaler engagement at a glance

Microsoft

BROAD ENGAGEMENT

Already builds data centers with CLT floors. Strategic interest in SUPERWOOD in the basis of design for future data centers; roof trusses leading to roofs and fast-to-build enclosures; immediate deployment of facades, interiors and fencing.

Meta

ENGAGED FOR OVER A YEAR

Broad engagement across the data center ecosystem. Facade applications and backplanes under way; plan to move into structural applications and racking over time.

Google

ENGAGEMENT JUST BEGUN

Initial focus on replacement of structural steel.

ALSO AT THE TABLE

Vertiv · Wooden Data Center · data center operators focused on security fencing

CONSTRUCTION PARTNERS

Enthusiastic support from some of the largest data center builders — HITT and Turner — alongside engineering partners Fast + Epp and Timber Engineering

Engagement stages as reported by InventWood, September 2026. Customer names used with permission.

CUSTOMER PROFILE

Microsoft *broad engagement*

ON THE PUBLIC RECORD

Two Northern Virginia datacenters built with cross-laminated timber floor panels on a steel frame — an estimated 35% embodied-carbon reduction vs. conventional steel construction and 65% vs. precast concrete. Design by Gensler; structural engineering by Thornton Tomasetti.

WORKING WITH INVENTWOOD

- **Strategic interest**
Incorporating SUPERWOOD into the basis of design for future data centers
- **Application areas**
Roof trusses, leading to roofs and fast-to-build building enclosures — developed with their principal architecture and engineering firms
- **Immediate deployment opportunities**
Building facades · biophilic interiors for office spaces · fencing for staff outdoor spaces · security fencing around critical outdoor infrastructure such as transformers



Concept rendering — SUPERWOOD security fencing around a transformer yard.

Source: Microsoft Source, "Microsoft builds first datacenters with wood to slash carbon emissions," Nov 2024; Thornton Tomasetti project page.

CUSTOMER PROFILE

Meta *engaged for over a year*

ON THE PUBLIC RECORD

Piloting mass timber for data center administrative buildings: first completed in 2025 at Aiken, South Carolina (DPR, SmartLam), with projects under way in Cheyenne, Wyoming and Montgomery, Alabama — about a 41% reduction in the embodied carbon of the materials substituted.

WORKING WITH INVENTWOOD

- **Broad engagement across every part of the data center ecosystem**

Conversations running for more than a year

- **Facade applications and backplanes under way**

Backplane certification path: UL 94 (yellow card)

- **Structural applications and racking next**

The plan is to move from skins into structure and racking over time



Concept rendering — SUPERWOOD equipment backplane in an electrical room.

Source: Meta Sustainability, "Meta pilots mass timber for more sustainable data center construction," 31 Jul 2025.

CUSTOMER PROFILE

Google *engagement just begun*

WORKING WITH INVENTWOOD

- **Initial focus: replacement of structural steel**

The structural-steel conversation starts at the qualification pathway — trusses first (no fire-resistance requirement), then assemblies

- **Early stage**

Scope and timeline to be defined with their design teams

A concrete path to first projects

1

Skins on one building or yard

Facades, interiors, louvers, staff-area or security fencing — installed from SuperMill One output, on a friendly site

2

Testing toward the standard the application needs

UL 94 for backplanes, STC/OITC for barriers, E119 and NFPA 285 for assemblies — scoped and co-funded with the customer

3

Basis-of-design work with the customer's architect and engineer

Structural enclosure and truss solutions that can be written into the standard campus design

4

Measure the carbon delta

A baseline bill-of-materials comparison and verification plan, so each project produces defensible numbers as the LCA completes



SuperMill One, Frederick, MD



Concept for SuperMill Two

Let's build what's *next*.



Alex Lau · CEO / Co-Founder · alex@inventwood.com

Lex Harris · Director, Capital Markets & IR · lex@inventwood.com